Core Java Study Plan Day 2

Programming Questions

1. Write a Java program that demonstrates the standard main method entry point and processes command-line arguments.

2. Write a Java program that demonstrates string immutability and the difference between string constant pool references and heap allocations (== vs .equals()).

3. Write a Java program to illustrate method overloading with multiple methods sharing the same name but different parameter lists.

4. Write a Java program to create a `Calculator` class with `add()`, `subtract()`, `multiply()`, and `divide()` methods.

5. Write a Java program to demonstrate object creation by creating an `Employee` class and creating two Employee objects.

6. Write a Java program to demonstrate encapsulation using a `BankAccount` class with a private `balance` variable.

7. Write a Java program to demonstrate inheritance using `Animal` and `Dog` classes.

8. Write a Java program to demonstrate method overriding using `Animal` and `Dog` classes.

9. Write a Java program to demonstrate runtime polymorphism using an `Animal` reference and a `Dog` object.

10. Write a Java program that simulates garbage collection eligibility by nullifying object references and requesting garbage collection via System.gc().

Puzzles

1. The Three Friends Puzzle

A, B, and C have different favorite colors: Red, Blue, and Green.

A does not like Red.

B does not like Green.

C likes Blue.

Question: What is A's favorite color?

A) Red

B) Blue

C) Green

D) Cannot be determined

2. The Age Puzzle

A father is 3 times as old as his son.

After 10 years, the father will be twice as old as his son.

Question: How old is the son now?

A) 5

B) 10

C) 15

D) 20

3. The Ranking Puzzle

Five students — A, B, C, D, and E — have different ranks.

A ranks higher than B.

C ranks lower than D.

B ranks higher than E.

D ranks higher than A.

Question: Which student must rank higher than A?

A) B

B) C

C) D

D) E

4. The Missing Number Puzzle

Find the next number:

3, 6, 11, 18, 27, 38, ?

A) 49

B) 51

C) 52

D) 54

5. The Clock Puzzle

A clock shows 3:00.

Question: What is the angle between the hour hand and the minute hand?

A) 60°

B) 90°

C) 120°

D) 180°